

Kestrel LED headlamp assembly moisture intrusion causing module configuration failure

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Area: bulletins

1 Condition

Some Kestrel vehicles may exhibit a body diagnostic trouble code B2477 (module configuration failure) stored in the body control module after replacement or service of the front LED headlamp assembly HLMP-7020. The condition typically presents as one or both headlamps remaining inoperative, intermittent flicker, or a configuration mismatch reported during scan tool communication. Moisture accumulation inside the HLMP-7020 housing corrupts the lamp driver's identification handshake with the BCM, which the module logs as B2477. For the full diagnostic flow refer to the Kestrel Body Service Manual :: Module Configuration Failure Diagnosis and the Kestrel Body Service Manual :: Body Diagnostic Trouble Code Reference .

2 Cause

The root cause is a degraded perimeter seal on the HLMP-7020 LED headlamp assembly that permits moisture intrusion into the integrated lamp driver electronics. As condensation bridges the driver's serial identification pins, the assembly fails to return a valid configuration signature, and the BCM sets B2477. This is distinct from a wiring or connector fault, so technicians should confirm the seal condition before condemning harness components. Background on the lighting circuit topology is available in the Kestrel Body Service Manual :: Lighting Sub-System and the Kestrel Body Service Manual :: Body Electrical System Overview .

3 Repair Procedure

Begin with PROC-HEADLAMP to remove and replace the affected HLMP-7020 headlamp assembly, inspecting the mounting surface and drain channels for trapped water before fitting the new unit. After installation, clear B2477 and perform the BCM relearn using PROC-BCM-CONFIG so the module captures the new assembly's configuration signature. Detailed replacement steps are in the Kestrel Body Service Manual :: Headlamp Assembly Replacement Procedure , and the configuration relearn is documented in the Kestrel Electrical Service Manual :: BCM Configuration Procedure . If the vehicle is equipped with the alternate body architecture, the equivalent relearn is the Vega Electrical Service Manual :: Body Control Module Configuration Procedure . Verify B2477 does not return after a key cycle and confirm both headlamps operate across all modes.